

BOTANY

1. C) Structurally resembles the substrate and competes for the active site. | Explanation: A competitive inhibitor has a similar chemical structure to the normal substrate and competes directly with it to bind at the enzyme's active site.
2. C) Magnesium | Explanation: Magnesium is the central atom in the porphyrin ring of the chlorophyll molecule. Without it, the plant cannot synthesize chlorophyll (causing chlorosis).
3. D) The author who first described the species | Explanation: According to binomial nomenclature rules, the abbreviated name at the end (Linn. for Linnaeus) indicates the taxonomist who first validly published the name.
4. B) Uptake of naked DNA fragments from the surrounding environment. | Explanation: Transformation is the genetic alteration of a cell resulting from the direct uptake and incorporation of exogenous genetic material from its surroundings.
5. C) Laminarin and mannitol | Explanation: Brown algae (Phaeophyceae) uniquely store their reserve food as the complex carbohydrates laminarin and mannitol.
6. C) Basidiocarp (fruiting body) | Explanation: In Basidiomycetes (mushrooms), the visible above-ground structure is the basidiocarp, which is a reproductive fruiting body bearing basidiospores.
7. B) The sporophyte is the dominant, independent phase. | Explanation: In Bryophytes, the main plant body is the gametophyte. In Pteridophytes (ferns), the main independent plant body is the diploid sporophyte (roots, stems, leaves).
8. C) Double fertilization and formation of triploid endosperm | Explanation: Double fertilization (resulting in a diploid zygote and a triploid primary endosperm nucleus) is a unique hallmark of angiosperms.
9. B) Rauwolfia serpentina | Explanation: Rauwolfia serpentina (Sarpagandha) is a medicinal plant well known for its alkaloids, primarily reserpine, which acts as an antihypertensive and tranquilizer.
10. A) Capsid | Explanation: The genetic material of a virus is enclosed in a protective protein shell called a capsid, which is made up of smaller subunits called capsomeres.
11. D) Brassicaceae | Explanation: Tetrastaminate stamens are a key identifying feature of the Brassicaceae (mustard) family.
12. B) Sulfur dioxide and Nitrogen oxides | Explanation: SO_2 and NO_x react with water vapor in the atmosphere to form sulfuric acid and nitric acid, which fall as acid rain.
13. C) Commensalism | Explanation: Commensalism is an interaction where one species benefits (the orchid gets sunlight/support) and the other is neither harmed nor benefited (the tree).
14. D) Carbon Dioxide (CO_2) | Explanation: While methane and nitrous oxide are potent, CO_2 is the most abundant and primarily responsible greenhouse gas driving human-induced climate change.
15. C) Benthos | Explanation: Benthic organisms (benthos) reside at the bottom of a body of water, in or on the substrate, as opposed to swimming freely (nekton) or drifting (plankton).
16. C) Bacterial: 70S; Plant: 80S. | Explanation: Prokaryotes (bacteria) contain 70S ribosomes, while the cytoplasm of eukaryotes (plants/animals) contains larger 80S ribosomes.
17. B) Synthesis and assembly of ribosomal RNA (rRNA) | Explanation: The nucleolus is a dense structure in the nucleus responsible for transcribing rRNA and assembling ribosomal subunits.
18. B) Lipid and steroid synthesis | Explanation: Smooth ER lacks ribosomes and is predominantly responsible for lipid synthesis, detoxification, and steroid hormone production.
19. C) G_0 phase | Explanation: The G_0 phase is a resting state where cells exit the active cell cycle and stop dividing, either temporarily or permanently.
20. B) Non-living cell inclusions like starch grains and crystals | Explanation: Ergastic substances are non-protoplasmic, non-living by-products of cell metabolism found in the cytoplasm or vacuoles, such as starch, tannins, and crystals (raphides).
21. B) Co-dominance | Explanation: In co-dominance, both alleles in a heterozygote (I^A and I^B) are fully and simultaneously expressed, creating a phenotype (AB blood) that shows traits of both parents.
22. B) One strand is original (parental) and one strand is newly synthesized. | Explanation: Semi-conservative replication ensures that each new double helix consists of one conserved parental strand acting as a template, paired with one newly built strand.

23. C) Chromosome 21 | Explanation: Down's syndrome is scientifically known as Trisomy 21, resulting from non-disjunction during meiosis which leaves three copies of chromosome 21 in the zygote.
24. B) Percentage of crossing over (recombination frequency) between them | Explanation: The closer two genes are, the lower the chance of a crossing-over event separating them. Therefore, recombination frequency is used to map genetic distance (1% recombination = 1 map unit / centimorgan).
25. C) It acts as an adaptor molecule, bringing specific amino acids to the mRNA-ribosome complex. | Explanation: tRNA reads the codons on the mRNA using its anticodon and fetches the corresponding amino acid to the ribosome to build the polypeptide chain.
26. B) RNA polymerase | Explanation: RNA polymerase is the enzyme that binds to the promoter region of DNA and synthesizes a complementary strand of RNA during transcription.
27. C) Endodermis | Explanation: The endodermis is the innermost layer of the cortex. Its Casparian strips (made of suberin) block apoplastic water flow, forcing water to pass through the cell membrane for filtration.
28. B) A well-developed, large central pith | Explanation: Monocot roots typically have a large, well-developed central pith and numerous xylem bundles (polyarch condition), whereas dicot roots have a very small or obsolete pith.
29. C) Sieve tube elements | Explanation: Sieve tube elements lose their nucleus and most organelles at maturity to create a clear channel for sap flow. Their functions are regulated by the adjacent companion cells.
30. B) Imbibition | Explanation: The initial rapid absorption of water by the dry, solid colloidal particles (like proteins and cellulose) of the raisin without forming a solution is called imbibition.
31. B) Mesophyll cells | Explanation: In C_4 plants (exhibiting Kranz anatomy), the first CO_2 fixation happens in the mesophyll cells via PEP carboxylase, while the Calvin cycle operates later in the bundle sheath cells.
32. B) Chloroplast, Peroxisome, Mitochondria | Explanation: The photorespiratory pathway (C_2 cycle) involves a complex interaction sequentially across three distinct cell organelles: Chloroplasts, Peroxisomes, and Mitochondria.
33. B) The expenditure of metabolic energy (ATP) | Explanation: Active transport moves mineral ions against their concentration gradient (from low in soil to high in root), which fundamentally requires energy supplied by ATP from respiration.
34. D) Ethylene | Explanation: Ethylene is a volatile, gaseous plant hormone widely known as the "ripening hormone" because it stimulates and accelerates the ripening process in climacteric fruits.
35. C) Glycolysis | Explanation: Glycolysis (the breakdown of glucose into pyruvate) occurs in the cytoplasm and requires no oxygen, making it the universal first step in both aerobic and anaerobic respiration.
36. B) Vegetative propagation | Explanation: Vegetative propagation is a form of asexual reproduction where new plants are grown from vegetative parts (stems, roots, leaves) without the involvement of seeds or spores.
37. B) Anemophily | Explanation: Anemophily refers to wind pollination. Entomophily is by insects, hydrophily by water, and ornithophily by birds.
38. B) Heterosis (Hybrid vigor) | Explanation: Heterosis or hybrid vigor refers to the tendency of a crossbred individual (hybrid) to show qualities superior to those of both outbred or inbred parents.
39. B) Generate thousands of genetically identical, virus-free plants. | Explanation: Meristems are usually devoid of viruses. Culturing them (micropropagation) allows for the rapid mass production of healthy, genetically identical clones (somaclones).
40. C) Beta-carotene (Pro-vitamin A) | Explanation: Golden Rice was engineered using genes from daffodils and bacteria to synthesize beta-carotene in the endosperm, which the human body converts into Vitamin A to prevent blindness.

ZOOLOGY

41. C) Analogous organs demonstrating convergent evolution. | Explanation: Analogous organs perform similar functions (e.g., flying) due to adapting to similar environmental pressures, but they do not share a common evolutionary origin. This evolutionary pattern, where unrelated organisms evolve similar traits, is called convergent evolution.
42. B) All organisms rely on the same universal genetic code (DNA/RNA) and fundamental metabolic pathways (like glycolysis). | Explanation: The fact that bacteria, plants, and humans all use the same base pairs (A, T, C, G) to code for the

same amino acids, and share core cellular processes (like using ATP for energy), is overwhelming evidence that all life descended from a single primordial lineage.

43. C) Differential survival and reproduction of individuals based on heritable variations best suited to their environment. | Explanation: Darwin's theory of Natural Selection centers on the idea that natural variation exists in populations. Individuals possessing traits that give them an advantage in their specific environment are more likely to survive and pass those advantageous traits to the next generation.

44. B) The coordinated beating of flagella on specialized collar cells called choanocytes. | Explanation: The inner walls of a sponge's canal system are lined with choanocytes (collar cells). These cells possess a single flagellum. The synchronized beating of millions of these flagella creates a continuous water current that draws water in through pores (ostia), traps food particles, and expels water through the osculum.

45. B) The division of the body into true linear segments, both externally and internally, allowing for coordinated movement. | Explanation: Metamerism is true segmentation. In annelids (like earthworms), the body is divided into a series of repeating segments (metameres) separated by septa. This segmentation allows for highly localized muscle contractions, making their burrowing movement highly efficient.

46. C) It is a ribbon-like structure covered in tiny chitinous teeth used to scrape food (like algae) off rocks. | Explanation: The radula is a uniquely molluscan feeding structure. It acts like a biological rasp or chainsaw, consisting of a flexible band armed with rows of microscopic teeth, allowing the organism to efficiently scrape or tear food.

47. C) The production of milk from modified sweat glands to nourish their young. | Explanation: While birds are also endothermic and many organisms have lungs or nerve cords, mammary glands (which produce milk for offspring) are the absolute defining and exclusive diagnostic feature of mammals.

48. B) It houses the blood vessels and nerves that supply the living bone cells (osteocytes). | Explanation: Bone is a living, highly vascular tissue. Because the calcified matrix is impermeable, blood vessels and nerves run lengthwise through the center of each osteon via the Haversian canal, ensuring all osteocytes receive nutrients and oxygen via connecting canaliculi.

49. B) The outermost layers become heavily packed with the tough, waterproof protein keratin. | Explanation: The epidermis is a 'keratinized' stratified squamous epithelium. As cells migrate to the surface, they die and become packed with keratin. This tough, dry, waterproof layer is perfect for protecting the body from mechanical injury, pathogens, and desiccation.

50. B) It is a loose matrix of fluid, varied fibers, and diverse cells that wraps organs, holding them in place while allowing for immune surveillance. | Explanation: Areolar tissue is a type of loose connective tissue. It is highly cellular and vascular, with a loose arrangement of collagen and elastin fibers. It acts as a universal packing tissue, binding skin to muscles, wrapping blood vessels, and housing immune cells (macrophages, mast cells).

51.C) A single, massive fat droplet occupies almost the entire cell, pushing the nucleus flat against the plasma membrane. | Explanation: White adipose tissue consists of 'unilocular' cells. As the cell stores more fat, a single large lipid droplet forms, expanding and squeezing the cytoplasm and nucleus to the very edge of the cell, giving it a 'signet ring' appearance under a microscope.

52. C) Amoeboid Trophozoite | Explanation: After a merozoite enters a red blood cell, it forms a central vacuole, pushing its nucleus to the side (the 'ring stage'). The parasite then grows, loses the vacuole, and develops pseudopodia, becoming the active, feeding 'amoeboid trophozoite' stage.

53.B) They exhibit protandry, meaning the testes mature and produce sperm much earlier than the ovaries produce eggs. | Explanation: Earthworms are protandrous. Their sperm matures before their own ova are ready. Therefore, an earthworm must mate with another earthworm to exchange sperm. They store the partner's sperm in spermathecae until their own eggs are mature, ensuring genetic diversity.

54. B) Digestive, Excretory, and Reproductive systems. | Explanation: Unlike mammals which have separate external openings, amphibians, reptiles, and birds possess a cloaca. It is a single terminal chamber that receives feces from the digestive tract, urine from the kidneys (excretory), and gametes (sperm/eggs) from the reproductive system before expelling them through the cloacal aperture.

55. A) Human have evolved to possess 12 pairs of cranial nerves.

56. C) They are made of chitin and anchor the worm's body segments to the soil during peristaltic locomotion. | Explanation: Setae are microscopic, S-shaped bristles composed of tough chitin, embedded in the body wall of most segments. When the longitudinal muscles contract, making the worm short and thick, the setae protrude to grip the soil, allowing the worm to push itself forward.

57. D) The Liver | Explanation: Immediately after infection, Plasmodium sporozoites leave the bloodstream and invade the hepatocytes (liver cells). This initiates the '*pre-erythrocytic cycle*'. They must multiply here for a week or two before breaking out to invade the red blood cells.

58. B) Secretion of digestive enzymes (like proteases) directly into the duodenum. | Explanation: The liver is a massive metabolic factory. It detoxifies, stores glycogen, synthesizes urea, and produces bile (an emulsifier). However, it does NOT produce digestive enzymes. The pancreas is responsible for secreting the major digestive enzymes (proteases, lipases, amylases) into the duodenum.

59. C) Medulla Oblongata and Pons | Explanation: The brainstem, specifically the medulla oblongata and the pons, houses the respiratory control centers. These centers monitor CO₂ and pH levels in the blood and cerebrospinal fluid, automatically adjusting the rate and depth of breathing via the phrenic nerve to the diaphragm.

60. B) It generates electrical impulses faster than any other autorhythmic tissue in the heart, driving the overall heart rate. | Explanation: Cardiac muscle cells are autorhythmic (they can generate their own action potentials). The SA node is the pacemaker because its cells reach the threshold potential and fire faster (about 70-80 times per minute) than the AV node or Purkinje fibers, thereby setting the rhythm for the entire heart.

61. B) It supplies oxygen and nutrients to the renal medulla without washing out the high osmotic concentration gradient established by the Loop of Henle. | Explanation: The renal medulla relies on a highly concentrated osmotic gradient to pull water out of the collecting ducts. The vasa recta (capillaries running parallel to the Loop of Henle) are highly permeable to water and solutes. They flow sluggishly, exchanging solutes and water in a way that nourishes the tissue without dissipating the crucial medullary gradient.

62. C) Stimulation of digestive secretions and increased intestinal peristalsis. | Explanation: The parasympathetic division governs the 'rest and digest' state. It conserves energy by slowing the heart rate, constricting pupils, and actively stimulating the digestive tract (increasing motility and gland secretion).

63. B) They are capable of continuous regeneration from basal stem cells throughout a person's life. | Explanation: Unlike most neurons in the central nervous system, which cannot divide once mature, olfactory receptor neurons are regularly replaced every 30 to 60 days. Basal cells in the olfactory epithelium act as stem cells to constantly generate new neurons, ensuring a sharp sense of smell is maintained.

64. B) Glucagon; it raises blood glucose by stimulating the liver to break down glycogen into glucose. | Explanation: The pancreas regulates blood sugar via two main hormones. Beta cells produce insulin (which lowers sugar). Alpha cells produce glucagon, which is released during fasting or exercise to raise blood sugar by stimulating glycogenolysis (breaking down glycogen) in the liver.

65. C) The trophoblast cells of the early developing embryo (blastocyst) and later the placenta. | Explanation: After fertilization, the developing blastocyst implants into the uterine wall. Its outer cell layer, the trophoblast (which will become the placenta), begins secreting hCG. This hormone signals the mother's corpus luteum to survive and keep producing progesterone, preventing menstruation and maintaining the pregnancy.

66. A) To act as a buffer and neutralize the highly acidic chyme entering the duodenum from the stomach. | Explanation: Chyme leaving the stomach is extremely acidic (pH ~2). Pancreatic enzymes (and the intestinal lining) cannot function in, and would be damaged by, this acid. Bicarbonate is an alkaline buffer that rapidly neutralizes the acid, raising the pH to around 7-8, creating the optimal environment for intestinal enzymes.

67. C) The Hypothalamus | Explanation: The hypothalamus is the master control center bridging the nervous and endocrine systems. It synthesizes releasing and inhibiting hormones that travel via a portal vein to control the anterior pituitary, and it physically extends neurons into the posterior pituitary to release ADH and oxytocin.
68. B) A concave lens, to diverge light rays before they enter the eye. | Explanation: In myopia, the eyeball is slightly too long, causing the natural lens to focus distant images prematurely (in front of the retina). A concave (diverging) lens is placed in front of the eye to spread the light rays slightly outward before they enter, pushing the focal point further back onto the retina.
69. D) Ampulla of oviduct | Explanation: Male sperm is ejaculated in female vagina with the help of penis during sexual intercourse. Sperm swims to oviduct through uterus and meets with ovum in ampulla of oviduct. Oviduct is also known as fallopian tube.
70. D) Neither A, B, nor Rh antigens | Explanation: Blood types are named for the antigens present. Type A has A, Type B has B, Type AB has both. Type O means the absence of A and B antigens. The 'negative' refers to the absence of the Rh factor (D antigen). Thus, an O- individual has none of these three antigens, making them the universal blood donor.
71. C) Cells | Explanation: Respiration can be simply defined as the process of breaking of complex food material into simpler form to release energy. This process occurs in mitochondria of cells. In lungs only gaseous exchange occurs.
72. B) The low pH (high acidity) of gastric juice in the stomach destroying ingested pathogens. | Explanation: Innate immunity is the first line of defense; it is present at birth and non-specific (it attacks any foreign body). The harsh, highly acidic environment of the stomach (pH 1.5 - 2) is a physiological barrier that indiscriminately kills the vast majority of bacteria ingested with food and water.
73. C) Two different strains of the virus infect the same host cell and reassort their genetic segments, creating a radically new viral subtype. | Explanation: Antigenic drift involves minor mutations, but antigenic shift is a major, sudden change. It occurs when a host (often a pig or bird) is infected by two different flu strains simultaneously. Their segmented RNA genomes mix (reassortment) inside the cell, producing a novel virus with brand new surface proteins (H and N) that the human population has no immunity against, often causing pandemics.
74. A) Lysozyme | Explanation: Lysozyme is an antimicrobial enzyme produced by animals as part of the innate immune system. It specifically breaks the chemical bonds in the peptidoglycan layer of bacterial cell walls, causing the bacteria to burst open (lyse).
75. B) Through direct contact with infected blood or body fluids (e.g., shared needles, sexual contact). | Explanation: Hepatitis B and C are bloodborne pathogens. They are most commonly transmitted through exposure to infective blood or body fluids, which occurs via sharing contaminated needles, unsterilized medical equipment, blood transfusions, or sexual contact.
76. C) The fungus is normally a harmless commensal organism in the body but causes disease only when the host's immune system is weakened or the normal bacterial flora is disrupted (e.g., by antibiotics). | Explanation: *Candida albicans* lives naturally in the human mouth, gut, and vagina without causing harm because the immune system and healthy bacteria keep it in check. It 'takes the opportunity' to overgrow and cause pathology (like thrush or yeast infections) when these defenses are compromised (e.g., in HIV/AIDS patients or after broad-spectrum antibiotic use).
77. C) The aggressive use of aerobic microorganisms in aeration tanks to biologically degrade dissolved organic matter. | Explanation: Primary treatment is strictly physical (screening and settling). Secondary treatment is biological. The effluent is moved into aeration tanks where oxygen is pumped in, encouraging aerobic bacteria and protozoa to rapidly consume the dissolved organic waste, drastically lowering the water's Biochemical Oxygen Demand (BOD).
78. B) Major Histocompatibility Complex (MHC) proteins | Explanation: MHC proteins (known as Human Leukocyte Antigens or HLA in humans) act as unique cellular fingerprints. Except for identical twins, every person's MHC profile is unique. If a transplanted organ has different MHC proteins, the recipient's Cytotoxic T-cells recognize them as foreign invaders and attack the tissue.
79. C) Wetlands (such as marshes, swamps, and lakes) critical for biodiversity, especially migratory waterbirds. | Explanation: The Ramsar Convention on Wetlands of International Importance was signed in Ramsar, Iran, in 1971. It provides the

framework for national action and international cooperation for the conservation and wise use of wetlands and their resources, recognizing them as highly productive and threatened ecosystems.

80. B) In-situ conservation | Explanation: In-situ (on-site) conservation involves protecting an endangered plant or animal species in its natural habitat by preserving the ecosystem itself. Ex-situ (off-site) conservation involves moving the organism out of its natural habitat into a controlled environment, such as a zoo, botanical garden, or seed bank.

CHEMISTRY

81. B) 2g of H_2 | Explanation: Moles of $H_2 = 2g / 2 g/mol = 1$ mole. $7g N_2 = 0.25$ mol. $16g NO_2 \approx 0.35$ mol. $16g O_2 = 0.5$ mol. More moles = more molecules.

82. C) 3, 3, 0, $+1/2$ | Explanation: The azimuthal quantum number 'l' must be strictly less than the principal quantum number 'n'. If $n=3$, it can only be 0, 1, or 2. $l=3$ is impossible.

83. D) Transition metals | Explanation: Transition metals possess incomplete d-subshells with multiple unpaired electrons, giving rise to pronounced paramagnetic behavior.

84. B) Linear | Explanation: The bond dipoles of $C=O$ cancel each other out perfectly only if the molecule has a linear, symmetrical geometry ($O=C=O$).

85. B) H^+ ions | Explanation: In an acidic medium, oxygen atoms are balanced by adding H_2O molecules to the deficient side, and the resulting hydrogen imbalance is fixed by adding H^+ ions to the other side.

86. D) Molar mass | Explanation: Graham's Law states that the rate of effusion or diffusion of a gas is inversely proportional to the square root of its molar mass (or density).

87. B) Absorb moisture and dissolve in it | Explanation: Deliquescent solids have such a high affinity for water that they absorb moisture from the air until they dissolve entirely to form an aqueous solution.

88. B) Proceed in the backward direction | Explanation: If $Q_c > K_c$, the ratio of products to reactants is higher than at equilibrium, so the system shifts left (backward) to consume products and reach equilibrium.

89. B) 1 Molal solution | Explanation: Molality (m) is defined as the number of moles of solute dissolved per kilogram of solvent.

90. B) Donates a proton | Explanation: According to Bronsted-Lowry, an acid is a proton donor. Once it donates its proton, the remaining species is its conjugate base.

91. C) Providing an alternative path with lower activation energy | Explanation: A catalyst provides a different reaction mechanism that has a lower activation energy barrier, allowing more molecules to react at a given temperature.

92. B) 0.0 V | Explanation: By convention, the Standard Hydrogen Electrode (SHE) is the reference point for electrochemical cells and is assigned a potential of exactly 0.00 V at all temperatures.

93. C) Neither absorbs nor releases heat | Explanation: An adiabatic process is defined as one in which no heat exchange ($q = 0$) occurs between the system and its surroundings.

94. B) Beta particle | Explanation: ^{14}C (atomic number 6) decays to ^{14}N (atomic number 7). To balance mass and charge, a beta particle (electron, mass 0, charge -1) must be emitted.

95. C) 3 | Explanation: Nitrogen gas (N_2) features a triple bond between the two nitrogen atoms to satisfy their octets, so the bond order is 3.

96. C) Zero | Explanation: Work ($W = P\Delta V$). In a vacuum, the external pressure (P) is zero. Therefore, expansion against zero pressure results in zero work done.

97. C) Greater than 7 | Explanation: At equivalence, the salt formed undergoes anionic hydrolysis (due to the weak acid's conjugate base), generating OH^- ions and making the $\text{pH} > 7$.
98. B) Has unpaired electrons (atomic state) | Explanation: Nascent hydrogen is freshly generated hydrogen in its atomic state (H). It is highly energetic and has an unpaired electron, making it a stronger reducing agent.
99. C) Nitrogen has a lone pair of electrons to donate | Explanation: A Lewis base is an electron pair donor. The nitrogen atom in ammonia (NH_3) has a lone pair of electrons that it can donate to form a coordinate bond.
100. B) Graphite | Explanation: Under standard conditions, graphite is the most thermodynamically stable allotrope of carbon, having the lowest free energy.
101. C) Zinc blende (Sphalerite) | Explanation: Zinc blende (Sphalerite) is primarily zinc sulfide (ZnS) and serves as the principal ore for extracting zinc.
102. B) Impurities are more soluble in the melt than in the solid metal | Explanation: Zone refining relies on the fact that impurities are more soluble in the molten state (melt) of a metal than in the solid state.
103. A) 3 parts HCl : 1 part HNO_3 | Explanation: Aqua regia, capable of dissolving noble metals like gold and platinum, is a 3:1 volumetric mixture of concentrated HCl and concentrated HNO_3 .
104. C) Hydrated iron(III) oxide | Explanation: Rust is hydrated iron(III) oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$), formed via an electrochemical process involving iron oxidation and oxygen reduction in the presence of moisture.
105. C) Boron | Explanation: Boron is a crucial micronutrient for plants (cell wall formation) but is generally not considered essential for animals.
106. B) Hg_2Cl_2 | Explanation: Calomel is mercury(I) chloride, written as Hg_2Cl_2 because it exists as a dimer containing a metal-metal bond.
107. B) SO_2 and NO_2 | Explanation: Acid rain is produced when sulfur dioxide (SO_2) and nitrogen oxides (NO_x) react with water and oxygen in the atmosphere to form sulfuric and nitric acids.
108. C) Functional groups | Explanation: Functional isomers have the same formula but different functional groups (e.g., ethanol and dimethyl ether are functional isomers with formula $\text{C}_2\text{H}_6\text{O}$).
109. B) Breaking large hydrocarbon molecules into smaller ones | Explanation: Cracking is the thermal or catalytic decomposition of large, heavy hydrocarbon molecules into smaller, more valuable fractions like gasoline and alkenes.
110. B) Asymmetrical alkenes | Explanation: Markovnikov's rule dictates that in electrophilic addition to an unsymmetrical alkene, the positive part of the reagent adds to the carbon with more hydrogen atoms.
111. A) Phosgene (highly poisonous) | Explanation: Chloroform slowly oxidizes in air and light to form the extremely toxic gas phosgene (COCl_2). 1% ethanol is added as a stabilizer.
112. B) Intermolecular hydrogen bonding | Explanation: Ethanol molecules form strong intermolecular hydrogen bonds, requiring more heat to separate them compared to dimethyl ether, which only has weak dipole forces.
113. B) Primary, secondary, and tertiary alcohols | Explanation: Lucas reagent (conc. HCl^+ anhydrous ZnCl_2) reacts at different rates with primary, secondary, and tertiary alcohols to form insoluble alkyl chlorides (cloudiness).
114. B) Ammoniacal silver nitrate | Explanation: Tollen's reagent is an ammoniacal solution of silver nitrate ($[\text{Ag}(\text{NH}_3)_2]^+$). It mildly oxidizes aldehydes, depositing a "silver mirror" on the test tube wall.
115. A) Alpha-hydrogen | Explanation: The aldol condensation requires an alpha-hydrogen atom, which is relatively acidic and can be removed by a base to form a nucleophilic enolate ion.

116. B) Concentrated acid catalyst | Explanation: Esterification is an equilibrium reaction that requires an acid catalyst (like conc. H_2SO_4) to activate the carbonyl carbon and remove the byproduct water.
117. B) Alkanes | Explanation: Clemmensen reduction uses zinc amalgam (Zn/Hg) and concentrated HCl to reduce the carbonyl group (C=O) of aldehydes and ketones completely to a methylene group (CH_2).
118. B) Aniline forms a salt with the Lewis acid catalyst (AlCl_3) | Explanation: Aniline is a Lewis base ($-\text{NH}_2$). It reacts with the Lewis acid catalyst AlCl_3 to form an insoluble salt, thereby deactivating the ring to further electrophilic attack.
119. B) Sn / HCl | Explanation: Tin (Sn) and hydrochloric acid (HCl) are commonly used in the lab to reduce nitro compounds to primary amines.
120. A) Primary alcohol | Explanation: Grignard reagents add to the carbonyl carbon. With formaldehyde, which has two H's attached to the carbonyl, the product after hydrolysis is a primary alcohol.
121. C) $-\text{O}-$ | Explanation: The ether functional group consists of an oxygen atom connected to two alkyl or aryl groups (R-O-R').
122. B) Acetylsalicylic acid | Explanation: Aspirin is synthesized by acetylating salicylic acid with acetic anhydride, producing acetylsalicylic acid.
123. D) Both B and C | Explanation: Carbocations are stabilized by electron-donating groups through positive inductive (+I) effect and resonance (+R) effect if conjugation is present.
124. C) Ethene | Explanation: At 170°C , conc. H_2SO_4 acts as a dehydrating agent, removing water from ethanol to form ethene via an elimination mechanism. (At 140°C , it forms ether).
125. C) Steady-state operation yielding constant product quality | Explanation: Continuous processing runs 24/7 without interruption, yielding a more consistent, steady-state output quality compared to batch processing which stops and starts.
126. C) DDT | Explanation: DDT (Dichlorodiphenyltrichloroethane) is a notorious, persistent organochlorine pesticide.
127. B) Sodium bicarbonate | Explanation: Baking soda is sodium bicarbonate (NaHCO_3), which releases CO_2 when heated or reacted with acid, causing dough to rise.
128. A) Ferric ferrocyanide | Explanation: In the Lassaigle test, nitrogen is converted to cyanide (CN^-). Treatment with iron salts produces the complex pigment Prussian blue, Ferric ferrocyanide ($\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$).
129. B) Vaporizes directly from solid to gas on heating | Explanation: Sublimation is used to separate a mixture where one component can transition directly from the solid to the gas phase (like iodine or camphor) without melting.
130. C) Be extremely pure and stable in air | Explanation: A primary standard must be highly pure, stable (not hygroscopic or efflorescent), have a high equivalent weight to minimize weighing errors, and react completely.

PHYSICS

131. Answer: A | Explanation: The exponent $\alpha z / k\text{BT}$ must be dimensionless. Since $k\text{BT}$ has dimensions of energy $[\text{ML}^2\text{T}^{-2}]$ and z is distance $[\text{L}]$, α must have dimensions of force $[\text{MLT}^{-2}]$. From $E = \alpha/\beta$, the dimension of β is $[\alpha]/[E] = [\text{MLT}^{-2}] / [\text{ML}^2\text{T}^{-2}] = [\text{L}^{-1}]$.
132. Answer: A | Explanation: The normal force is $N = mg \cos\theta + F \sin\theta$. To prevent sliding down, friction $f = \mu_s N$ acts upwards along the incline. Balancing forces along the incline gives $mg \sin\theta = F \cos\theta + \mu_s (mg \cos\theta + F \sin\theta)$. Solving for F yields $mg(\sin\theta - \mu_s \cos\theta) / (\cos\theta + \mu_s \sin\theta)$.
133. Answer: A | Explanation: According to Poiseuille's law, fluid resistance is $R_f = 8\eta L / (\pi r^4)$. For tube 1, $R_1 \propto L/R^4$. For tube 2, $R_2 \propto 2L/(2R)^4 = 2L/(16R^4) = (1/8)(L/R^4)$. In series, flow rate is constant, so pressure drop is proportional to resistance. $R_1 / R_2 = 8 : 1$.

134. Answer: A | Explanation: Total energy is Mgh . Kinetic energy is $K_{\text{trans}} + K_{\text{rot}} = (1/2)Mv^2 + (1/2)I\omega^2$. For a solid sphere, $I = (2/5)MR^2$, making $K_{\text{rot}} = (2/5)K_{\text{trans}}$. Thus, total energy is $(7/5)K_{\text{trans}}$, meaning $K_{\text{trans}} = (5/7)Mgh$.

135. Answer: A | Explanation: Initial potential energy is $U_i = -GMm / R$ or $-mgR$. Final potential energy at $h = R/2$ is $U_f = -GMm / (R + R/2) = -2GMm / (3R) = -(2/3)mgR$. Work done is $U_f - U_i = -(2/3)mgR - (-mgR) = (1/3)mgR$.

136. Answer: A | Explanation: Energy density is $u = (1/2) \times \text{stress} \times \text{strain}$. Since stress $= F/A$ and strain $= \text{stress}/Y = F/(AY)$, the energy density is $u = (1/2) (F/A) (F/AY) = F^2 / (2A^2Y)$.

137. Answer: A | Explanation: When displaced by x , one spring compresses and pushes back with force k_1x , while the other stretches and pulls back with k_2x . Both forces act in the same direction toward equilibrium. Net restoring force is $-(k_1 + k_2)x$, making the effective constant $k_1 + k_2$.

138. Answer: A | Explanation: At the highest point, the vertical velocity is zero, so the total velocity is just the horizontal component, $v = u \cos\theta$. The only acceleration is gravity g , acting perpendicularly. Using centripetal acceleration $a_c = v^2/R$, we get $g = (u \cos\theta)^2 / R$, so $R = (u^2 \cos^2\theta) / g$.

139. Answer: A | Explanation: The velocity of efflux is driven by hydrostatic pressure, which depends on effective gravity g_{eff} . In a freely falling reference frame, the effective gravity is $g_{\text{eff}} = g - a = g - g = 0$. Without effective gravity, there is no pressure gradient to force the liquid out.

140. Answer: A | Explanation: From conservation laws, the velocity of the second mass after an elastic collision is $v_2 = 2mv / (m+M)$. Its kinetic energy is $K_2 = (1/2)Mv_2^2 = (1/2)M(2mv / (m+M))^2$. Dividing this by the initial kinetic energy $K_1 = (1/2)mv^2$ yields the fraction $4mM / (m+M)^2$.

141. B) Molecular Motion Ceases | Explanation: Absolute zero (0 K) is the lowest theoretical temperature where all thermodynamic and kinetic motion of the molecules completely ceases.

142. B) $2K_1K_2/(K_1 + K_2)$ | Explanation: For two equivalent rods connected in series, the total thermal resistance is $R = R_1 + R_2$. Substituting $R = l / (KA)$, we get $2l / (K_{\text{eq}} A) = l / (K_1 A) + l / (K_2 A)$, which simplifies to $K_{\text{eq}} = 2K_1K_2 / (K_1 + K_2)$.

143. C) Conservation of Energy | Explanation: The first law of thermodynamics, represented by $\Delta Q = \Delta U + \Delta W$, is a direct statement of the universal law of conservation of energy applied to thermal systems.

144. D) Entropy (Heat) | Explanation: In an adiabatic process, there is no heat exchange between the system and its surroundings ($\Delta Q = 0$). A reversible adiabatic process is isentropic, meaning the entropy (S) remains constant.

145. B) 80 cal/g | Explanation: The latent heat of fusion of ice is the amount of heat required to change 1 gram of ice into water at 0°C without changing its temperature, which is equal to 80 cal/g (or approximately 336 J/g).

146. A) Absorbs all incident radiation | Explanation: An ideal perfect black body is a theoretical object that completely absorbs all wavelengths of electromagnetic and thermal radiation incident upon it, without reflecting or transmitting any.

147. C) $C_p - C_v = R$ | Explanation: Mayer's formula establishes the relation between the specific heat capacity at constant pressure (C_p) and constant volume (C_v) for an ideal gas, given by $C_p - C_v = R$.

148. B) Frequency | Explanation: Pitch is a subjective auditory sensation that depends directly on the objective physical property of the sound wave's frequency. Higher frequency corresponds to a higher pitch.

149. C) Interference | Explanation: Beats are regular variations in sound intensity produced by the superposition or interference of two sound waves traveling in the same direction with slightly different frequencies.

150. C) Frequency remains constant | Explanation: Frequency is a characteristic property of the source producing the wave and remains unchanged when light crosses an optical boundary. However, velocity and wavelength decrease in a denser medium.

151. B) 20 cm | Explanation: Cutting a convex lens symmetrically along its principal axis does not alter the radius of curvature of its refracting surfaces. Therefore, the focal length of each individual half remains 20 cm.

152. A) Dispersion | Explanation: Dispersion is the physical phenomenon where white light splits into its constituent spectrum of colors when passing through a medium like a prism, due to different wavelengths refracting at different angles.
153. C) Total Internal Reflection | Explanation: Optical fibers function by transmitting light pulses down a glass or plastic strand via successive reflections off the core-cladding boundary, acting on the principle of Total Internal Reflection.
154. B) $\mu = \tan i_p$ | Explanation: Brewster's law states that when unpolarized light is incident at the polarizing angle (i_p), the reflected light is completely polarized, and the refractive index of the medium is given by $\mu = \tan(i_p)$.
155. C) Remains unchanged | Explanation: The fringe width formula in Young's Double Slit Experiment is $\beta = \lambda D/d$. Moving the source closer to the slits alters the illumination intensity but does not change the distance to the screen (D) or slit separation (d), leaving β unchanged.
156. C) Its material | Explanation: Specific resistance (or resistivity) is an intrinsic property of a material that depends solely on the nature of the material itself and its temperature. It is entirely independent of geometric dimensions like length or cross-sectional area.
157. B) Low resistance in parallel (shunt) | Explanation: A galvanometer is converted into an ammeter by bypassing the majority of the circuit current around the sensitive meter. This is achieved by connecting a very low resistance, called a shunt, in parallel with it.
158. B) Zero | Explanation: Kirchhoff's Voltage Law (KVL) is based on the conservation of energy and states that the algebraic sum of all electromotive forces (EMFs) and potential drops around any closed loop in a circuit must equal zero.
159. A) 00 | Explanation: At the Earth's magnetic equator, the magnetic field lines run perfectly horizontal and parallel to the surface. As a result, a magnetic needle experiences no vertical tilt, making the angle of dip 0° .
160. B) Weber | Explanation: The SI unit of magnetic flux (total magnetic field passing through a given area) is the Weber (Wb). Tesla is the unit for magnetic flux density.
161. A) Bvl | Explanation: When a straight conductor of length l cuts across a uniform magnetic field B with a perpendicular velocity v , the induced motional electromotive force (emf) is calculated using $E = Bvl$.
162. B) Magnetic field | Explanation: An inductor is a passive component that stores electrical energy temporarily in the form of an inductive magnetic field when current passes through its coils ($E = \frac{1}{2} \times L \times I^2$).
163. C) 1 | Explanation: The power factor is given by $\cos \phi$, where ϕ is the phase angle between voltage and current. In a purely resistive circuit, voltage and current are perfectly in phase ($\phi = 0^\circ$), so $\cos(0^\circ) = 1$.
164. C) Rectifier | Explanation: A rectifier is an electronic device, typically configured with semiconductor diodes, that permits current to flow in only one direction, effectively converting alternating current (AC) into direct current (DC).
165. C) Central force | Explanation: The Coulomb force between static charges acts strictly along the line connecting the centers of the two charges and its magnitude depends only on the distance between them, which defines a central force.
166. C) Zero | Explanation: By definition, the electric potential is equal at all points across an equipotential surface, meaning $\Delta V = 0$. The work done moving a charge is $W = q \times \Delta V = q \times 0 = 0$.
167. B) $6\mu F$ | Explanation: When capacitors are connected in a parallel configuration, their equivalent capacitance is the direct algebraic sum of their individual capacitances: $C_{eq} = C_1 + C_2 = 2\mu F + 4\mu F = 6\mu F$.
168. D) Infinite | Explanation: Inside an ideal conductor or metal, mobile free electrons realign to completely cancel out external static electric fields, meaning the internal net field is zero. This property corresponds to an infinite relative permittivity (dielectric constant).

169. Answer: B | Explanation: When falling freely, the downward weight is balanced by viscous drag: $mg = 6\pi\eta r v_1$. When rising under the electric field, the upward electric force balances both weight and downward viscous drag: $qE = mg + 6\pi\eta r v_2$. From the first equation, $6\pi\eta r = mg / v_1$. Substitute this into the second: $qE = mg + (mg / v_1) v_2 = mg (1 + v_2 / v_1) = mg (v_1 + v_2) / v_1$. Therefore, $q = mg (v_1 + v_2) / (Ev_1)$.

170. C) Frequency of light | Explanation: Einstein's photoelectric equation is $K_{\max} = h\nu - \Phi$. This demonstrates that the maximum kinetic energy of the emitted photoelectrons increases in a direct linear relationship with the frequency (ν) of the incident light.

171. B) Visible region | Explanation: The Balmer series of the hydrogen emission spectrum corresponds to electronic transitions dropping down to the $n = 2$ energy level, emitting photon wavelengths that fall squarely within the human visible light region.

172. Answer: C | Explanation: According to Hubble's law, velocity of recession $v = H_0 \times d$. Therefore, distance $d = v / H_0 = (3 \times 10^4) / 75 = 30000 / 75 = 400$ Mpc.

173. B) $\tau = 1/\lambda$ | Explanation: The mean life (or average life expectancy, τ) of a radioactive atom is defined as the statistical reciprocal of its characteristic radioactive decay constant (λ).

174. B) Cadmium | Explanation: Control rods in a nuclear reactor regulate the chain reaction by absorbing excess thermal neutrons. Materials like Cadmium or Boron are utilized because they possess exceptionally high neutron absorption cross-sections.

175. A) 9×10^{16} J | Explanation: According to Einstein's mass-energy equivalence principle, $E = mc^2$. Substituting the values gives $E = 9 \times 10^{16}$ Joules.

176. C) Voltage regulator | Explanation: A Zener diode is heavily doped to allow it to operate safely in its reverse breakdown region. Because it maintains a stable breakdown voltage across a wide range of currents, it is primarily used as a voltage regulator.

177. A) OR gate | Explanation: The boolean expression $A + B$ represents the operation of an OR gate.

178. C) An insulator | Explanation: At absolute zero (0 K), semiconductor valence electrons lack the thermal energy required to break out of covalent bonds and jump into the conduction band. Therefore, pure silicon possesses zero free charge carriers and acts as a perfect insulator.

179. B) Current operated devices | Explanation: Bipolar Junction Transistors (BJTs) are fundamentally current-controlled or current-operated devices, where a small base current (I_b) directly controls a significantly larger collector current (I_c).

180. A) Position and momentum | Explanation: Heisenberg's uncertainty principle sets a fundamental physical limit on measurement, stating that the complementary variables of position (Δx) and momentum (Δp) cannot be simultaneously determined with absolute precision ($\Delta x \Delta p \geq h / 4\pi$).

MAT

181. B) | Explanation: OBFUSCATE removes CLARITY. MOLLIFY removes ANGER.

182. C) | Explanation: Since some trunks are zinds, and absolutely NO zinds are frelks, those specific trunks that are zinds cannot be frelks.

183. D) | Explanation: Ephemeral, Evanescent, and Transient mean fleeting. Pertinacious means stubborn/persistent.

184. B) | Explanation: If blips are entirely within blops, and blops do not overlap with gloops at all, no blips can be gloops.

185. D) | Explanation: If drivers prefer to pay rather than take local roads, the predicted local congestion (the flaw) won't occur.

186. A) | Explanation: Pattern: $(x \times 3) - 1 = 365$.

187. D) | Explanation: Combined fill in first 6 hours = $\frac{1}{6}$ of tank. A finishes the remaining $\frac{5}{6}$ in 10 hours. Total = 16.

188. B) | Explanation: $(10C1 \times {}^5C_2) / {}^{15}C_3 = 100/455 = 20/91$.

189. B) | Explanation: Difference of 18 means digit difference is 2 ($a-b=2$). Sum is 10 ($a+b=10$). $a=6$, $b=4$. Age = 64.

190. A) | Explanation: Winner gets 54%, loser 46%. Diff is 8% of valid votes ($0.81x$). $0.08 * 0.81x = 1620 \rightarrow x = 25,000$.

191. A) | Explanation: First letters +2, +3, +4, +5. Second letters -2, -3, -4, -5. Result: O, L.

192. A) | Explanation: Pattern: $(x \times n) + n$. $(136 \times 5) + 5 = 685$.

193. B) | Explanation: Sequence must be B-A-C-E-D. C is strictly in the middle.

194. A) | Explanation: Edges swapped, middle reversed and shifted by +1. MEDICINE $\rightarrow$ E + OJDJEF + M.

195. D) | Explanation: 25th of the month. Since the 3rd is Wednesday, 24th is Wednesday, making 25th Thursday.

196. B) | Explanation: The logic across the rows is consistent. Column 1 establishes a base geometric shape. Column 2 adds a horizontal bisecting line. Column 3 adds a vertical intersecting line (forming a cross). Therefore, the missing shape in Row 3 (which begins with a Triangle) must be a Triangle containing both a horizontal and vertical crossline.

197. D) | Explanation: Figures A, B, and C all feature shapes that have been intersected by exactly two straight lines, creating 4 internal symmetrical sectors. Figure D, the triangle, is intersected by only one vertical line, creating 2 sectors, breaking the visual pattern.

198. A) | Explanation: Rule 1 (Rotation): The arrow rotates 90 degrees clockwise each step (12 $\rightarrow$ 3 $\rightarrow$ 6 $\rightarrow$ 9 o'clock / Left). Rule 2 (Element addition): An arrowhead is added each step (1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4). Rule 3 (Shading): The arrowheads alternate in fill color (Black $\rightarrow$ Black/White $\rightarrow$ Black/White/Black $\rightarrow$ Black/White/Black/White).

199. A)

200. A)